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ETHANOL RESISTANT EXTENDED RELEASE MATRIX TABLETS WITH COMPRITOL® 888 ATO

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INTRODUCTION

Accidental dose dumping induced by the concomitant intake of alcoholic beverages with a drug product is mainly linked to the uncontrolled dissolution of the functional excipient in ethanol[1,2]. Dose dumping becomes a serious safety concern with extended release products, which contain higher drug concentrations compared to immediate release dosage forms. Compritol® 888 ATO is a lipid matrix agent used in sustained release tablets. It is insoluble in ethanol and exhibits interesting features to prevent alcohol related dose dumping.

EXPERIMENTAL METHODS

Compritol® 888 ATO (glyceryl behenate NF), drug and excipients (Table 1) were blended using a Turbula at 90 rpm for 10 min. The powder blend was lubricated with magnesium stearate at 90 rpm for 1 min prior to compression with flat tooling at 18 kN. 100 mg theophylline tablets were obtained using a single station press (Korsch EK0, 333 mg total weight, 10 mm diameter). 500 mg niacin tablets were prepared with a multi station press (Riva Piccola, 1000 mg total weight, 12 mm diameter). Drug release from tablets was measured in 0.1N HCl and 0.1N HCl: 40% ethanol for 2h at 37°C (Sotax AT7), referring to the FDA worst case guidance. Similarity of drug release in both media was concluded when $50 \le f2 \le 100$. The wettability of tablets was quantified by measuring the contact angle of 0.1N HCl and 0.1N HCl: 40% ethanol to the tablet surface. 15 µL of liquid was dropped onto the flat tablet surface using a goniometer (R GBX 01) equipped with a Nikon camera. The mean value of the left and the right contact angle was calculated.

Table 1: Composition of sustained release matrix tablets.

Ingredients, % w/w	Formulation F1	Formulation F2	Formulation F3	Formulation F4
Theophylline	30	30	-	-
Niacin	-	-	63	63
Compritol® 888 ATO	19.5	19.5	20	20
Avicel PH101 (MCC)	50	-	-	-
Flowlac 100 (lactose)	-	50	-	-
Emcompress (DCP)	-	-	15	-
Ethocel 10 FP (EC)			-	15
Aerosil Pharma 200			1.5	1.5
Magnesium stearate	0.5	0.5	0.5	0.5

RESULTS AND DISCUSSION

Compritol® 888 ATO does not dissolve in water or ethanol. Excipient-related dose dumping in hydro-alcoholic media is therefore absent. However, the solubility of the drug can significantly change. Since drug release from Compritol® 888 tablets is driven by diffusion this change in drug solubility will likely affect the release kinetics.

Figure 1 shows the effect of hydro-alcoholic media on theophylline solubility and the consequence on tablet wettability and drug release from Compritol® 888 tablets containing MCC as filler (F1). Theophylline is threefold more soluble in 40% ethanol compared to 0.1N HCl (Figure 1a). This results in improved tablet wetting (Figure 1b). Consequently theophylline release is faster in 0.1N HCl: 40% ethanol than in 0.1N HCl (f2 = 33.5, Figure 1c). Such undesired rapid drug release in hydro-alcoholic fluid can be prevented by using a filler that is less soluble in ethanol than in HCl: e.g. lactose. When replacing MCC by lactose (F2), tablet wetting in 40% ethanol is reduced and is more similar to the tablet wettability in 0.1N HCl (Figure 1d). Theophylline release in the presence of lactose is hence unaffected in both media with f2 = 73.5 although drug solubility is threefold increased (Figure 1e).

The opposite occurs with niacin (vitamin B3) which exhibits twofold less solubility in 0.1N HCl: 40% ethanol compared with 0.1N HCl (Figure 2a). This results in slower release kinetics when DCP is the filler (f2 = 37, Figure 2b). Similar to the first part of this study the release kinetics can be adjusted by adding an ethanol-soluble filler such as ethylcellulose to the formulation, which counterbalances the reduced solubility of niacin (Figure 2c).

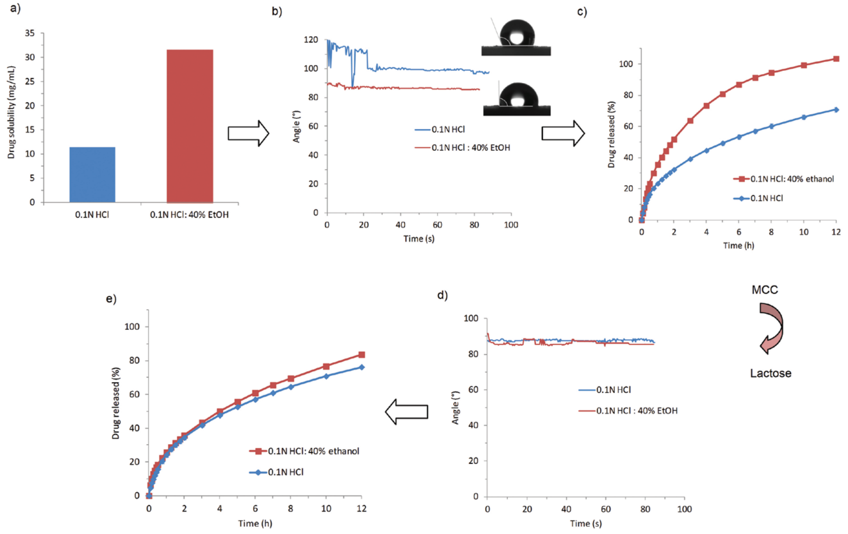

Figure 1: Effect of drug solubility on matrix wettability and drug release kinetics in 0.1N HCl and 0.1N HCl: 40% ethanol from tablets produced by direct compression with Compritol® 888 ATO, MCC (F1) and lactose (F2).

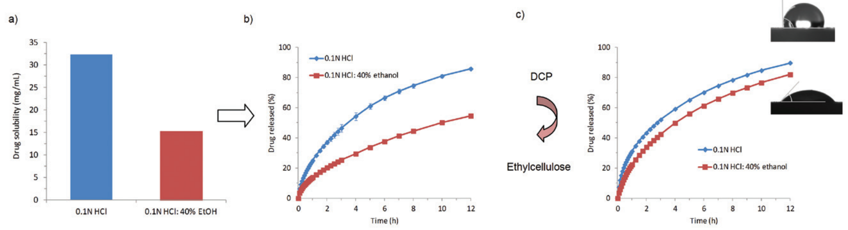

Figure 2: Effect of hydro-alcoholic media on: a) niacin solubility, b) niacin release from Compritol® 888: DCP tablets (F3) and c) niacin release from Compritol® 888: ethylcellulose tablets (F4).

CONCLUSION

For certain drug compounds the prevention of alcohol related dose dumping is pre-requisite. This needs to be addressed during the formulation stage of drug development. More and more compounds are coming under scrutiny and application of the FDA guidelines for testing for resistance to alcohol should be considered during drug development. The use of Compritol® 888 ATO with appropriate diluents provides a practical and robust solution for the formulation of drugs whose solubility hence dissolution from extended release tablets can be affected by alcohol.

REFERENCES

[1] Walden, M. et al. Drug Dev Ind Pharm. 2007, 33, 1101–1111. [2] Fadda, H. et al. Int. J. Pharm. 2008, 360, 171-76.